

The torque controlled 3R planar robot in Fig. 1 moves on a horizontal plane, performing a two-dimensional trajectory task with its end-effector. The links have equal length l and equal uniformly distributed mass m , with barycentric inertia $I_c = ml^2/12$. While at rest in the configuration $q = (\pi/4, -\pi/2, \pi/2)$ [rad], the end-effector should accelerate with $\ddot{p}_d = (1, 0)$ [m/s²] (see in figure).

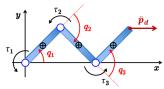


Figure 1: A 3R planar robot with equal links.

- Determine, in a parametric way with respect to l and m , the torques $\tau_i \in \mathbb{R}^3$, for $i = A, B, C$, that realize instantaneously the following objectives:
- τ_A minimizes the squared norm of the joint accelerations $H_A = \frac{1}{2} \|\ddot{q}\|^2$;
 - τ_B minimizes the squared norm of the absolute joint accelerations $H_B = \frac{1}{2} \|\ddot{q}_a\|^2$, where $\ddot{q}_{a,i} = \sum_{j=1}^i \ddot{q}_j$, $i = 1, 2, 3$;
 - τ_C minimizes the squared norm of the inertia-weighted joint accelerations $H_C = \frac{1}{2} \ddot{q}^T M(q) \ddot{q}$.
- Comment on the obtained results in terms of the control efforts at the joint level.

$$P_{c1} = \begin{cases} l_{1/2} c_1 \\ l_{1/2} s_1 \end{cases} \quad P_{c2} = \begin{cases} l c_1 + \frac{l}{2} c_{12} \\ l s_1 + \frac{l}{2} s_{12} \end{cases} \quad P_{c3} = \begin{cases} l(c_1 + c_2) + \frac{l}{2} c_{123} \\ l(s_1 + s_2) + \frac{l}{2} s_{123} \end{cases}$$

$$V_{c1} = \begin{cases} -\frac{l}{2} s_1 \dot{q}_1 \\ \frac{l}{2} c_1 \dot{q}_1 \end{cases} \quad V_{c2} = \begin{cases} -l s_1 \dot{q}_1 - \frac{l}{2} s_{12} (\dot{q}_1 + \dot{q}_2) \\ l c_1 \dot{q}_1 + \frac{l}{2} c_{12} (\dot{q}_1 + \dot{q}_2) \end{cases} \quad V_{c3} = \begin{cases} -l(s_1 \dot{q}_1 + s_{12} (\dot{q}_1 + \dot{q}_2)) - \frac{l}{2} s_{123} (\dot{q}_1 + \dot{q}_2 + \dot{q}_3) \\ l(c_1 \dot{q}_1 + c_{12} (\dot{q}_1 + \dot{q}_2)) + \frac{l}{2} c_{123} (\dot{q}_1 + \dot{q}_2 + \dot{q}_3) \end{cases}$$

• At rest: $\ddot{q} = J^\# \ddot{p}_d$ hence we have to execute $J^\# \ddot{p}_d$ in order to minimize $\frac{1}{2} \|\ddot{q}_d\|^2 \Rightarrow \tau = M J^\# \ddot{p}_d + \cancel{c}$. We have to explicit $J^\#$, M (not c since is at rest)

$$K(q) = \begin{cases} l(c_1 + c_{12} + c_{123}) \\ l(s_1 + s_{12} + s_{123}) \end{cases} \Rightarrow J = l \begin{pmatrix} -(s_1 + s_{12} + s_{123}) & -(s_{12} + s_{123}) & -s_{123} \\ c_1 + c_{12} + c_{123} & c_{12} + c_{123} & c_{123} \end{pmatrix} \Rightarrow J(\bar{q}) = l \begin{pmatrix} -1 - \sqrt{2} & -1 - \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \sqrt{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\ddot{q}_d = J^\# \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \text{ Now i compute } M.$$

$$T = \frac{1}{2} I_c (3\dot{q}_1^2 + 2\dot{q}_2^2 + \dot{q}_3^2) + \frac{1}{2} M \left[\frac{l^2}{4} \dot{q}_1^2 + l^2 \dot{q}_1^2 + \frac{l^2}{4} (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1 \dot{q}_2) + 2 \frac{l^2}{2} (\dot{q}_1^2 + \dot{q}_1 \dot{q}_2) C_2 + \right. \\ \left. l^2 \dot{q}_1^2 + l^2 (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1 \dot{q}_2) + 2l^2 (\dot{q}_1^2 + \dot{q}_1 \dot{q}_2) C_2 + \frac{l^2}{4} (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2 + 2\dot{q}_1 \dot{q}_2 + 2\dot{q}_1 \dot{q}_3 + 2\dot{q}_2 \dot{q}_3) + \right. \\ \left. 2 \frac{l^2}{2} (\dot{q}_1^2 + \dot{q}_1 \dot{q}_2 + \dot{q}_1 \dot{q}_3) C_{23} + 2 \frac{l^2}{2} (\dot{q}_1^2 + 2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2 + \dot{q}_2 \dot{q}_3 + \dot{q}_1 \dot{q}_3) C_3 \right] =$$

$$M_{11} = m l^2 (4 + 3C_2 + C_{23} + C_3) \quad M_{22} = m l^2 \left(\frac{5}{3} + C_3 \right) \quad M_{33} = m l^2 \frac{1}{3}$$

$$M_{12} = m l^2 \left(\frac{3}{2} + \frac{3}{2} C_2 + \frac{1}{2} C_{23} + C_3 \right) \quad M_{23} = m l^2 \left(\frac{1}{4} + \frac{1}{2} C_3 \right) \quad M_{13} = m l^2 \left(\frac{1}{4} + \frac{1}{2} C_{23} + \frac{1}{2} C_3 \right)$$

$$M = m l^2 \begin{bmatrix} 4 + 3C_2 + C_3 + C_{23} & \frac{3}{2} + \frac{3}{2} C_2 + C_3 + \frac{1}{2} C_{23} & \frac{1}{4} + \frac{1}{2} C_3 + \frac{1}{2} C_{23} \\ \frac{3}{2} + \frac{3}{2} C_2 + C_3 + \frac{1}{2} C_{23} & \frac{5}{3} + C_3 & \frac{1}{4} + \frac{1}{2} C_3 \\ \frac{1}{4} + \frac{1}{2} C_3 + \frac{1}{2} C_{23} & \frac{1}{4} + \frac{1}{2} C_3 & \frac{1}{3} \end{bmatrix} = m l^2 \begin{pmatrix} 5 & 2 & 3/4 \\ 2 & 5/3 & 1/4 \\ 3/4 & 1/4 & 1/3 \end{pmatrix}$$

$$\tau = M \ddot{q}_d = m l \begin{pmatrix} \frac{1}{4} + \frac{1}{2} C_3 + \frac{1}{2} C_{23} - \left(\frac{3}{2} + \frac{3}{2} C_2 + C_3 + \frac{1}{2} C_{23} \right) \\ \frac{1}{4} + \frac{1}{2} C_3 - (5/3 + C_3) \\ \frac{1}{3} - \left(\frac{1}{4} + \frac{1}{2} C_3 \right) \end{pmatrix} = \frac{m l}{2} \begin{bmatrix} -\frac{1}{6} - 3C_2 - C_3 \\ -\frac{13}{6} - C_3 \\ \frac{1}{6} - C_3 \end{bmatrix} =$$

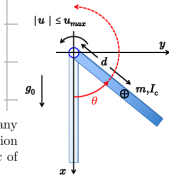
Now consider $H_B = \frac{1}{2} (\ddot{q}_1 \quad \ddot{q}_1 + \ddot{q}_2 \quad \ddot{q}_1 + \ddot{q}_2 + \ddot{q}_3)^T (\ddot{q}_1 \quad \ddot{q}_1 + \ddot{q}_2 \quad \ddot{q}_1 + \ddot{q}_2 + \ddot{q}_3) = 5\ddot{q}_1^2 + 3\ddot{q}_2^2 + \ddot{q}_3^2 + 6\ddot{q}_1 \ddot{q}_2 + 2\ddot{q}_2 \ddot{q}_3 + 2\ddot{q}_1 \ddot{q}_3$

$$= \ddot{q}^T \underbrace{\begin{pmatrix} 5 & 6 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{pmatrix}}_B \ddot{q} \quad \text{so we choose the pseudo inverse weighted by } B \quad \tau = M J_B^\# \ddot{p}_d$$

At the end, to minimize $\ddot{q}^T M \ddot{q}$ we use $z = M J_M^* \ddot{p}_d = J^T (J M^T J^T)^{-1} \ddot{p}_d$

Exercise 2

Consider the single link under gravity in Fig. 2, with all dynamic parameters specified therein. The link should perform a rest-to-rest motion from $\theta(0) = 0$ to $\theta(T) = \pi$ (a swing-up maneuver, counterclockwise), by following a cubic polynomial interpolating trajectory $\theta_d(t)$ under the torque bound $|u| \leq u_{max}$. Suppose that the maximum available torque is large enough to sustain gravity in any configuration, typically with some extra torque left for dynamic motion.



Determine the expression of the minimum time $T = T_{min}$ for performing the task, discussing any assumption that you may introduce. Furthermore, suppose that, with a time $T = 1$ s, the motion is unfeasible for a given set of data. What will be the minimum uniform time scaling factor k of the original trajectory that allows to execute the task in a feasible way?

The model is $(I_c + m d^2) \ddot{\theta} + m g_0 \sin \theta = u \Rightarrow$

$\ddot{\theta} = (I_c + m d^2)^{-1} (u - m g_0 \sin \theta)$ the trajectory is $\theta_d(t) = at^3 + bt^2 + ct + d$

$$\theta_d(0) = at^3 + bt^2 + ct + d = 0$$

$$\theta_d(T) = aT^3 + bT^2 = \pi$$

$$\dot{\theta}_d(0) = c = 0$$

$$\dot{\theta}_d(T) = 3aT^2 + 2bT = 0$$

$$\Rightarrow \begin{cases} aT^3 + bT^2 = \pi \\ 3aT^2 + 2bT = 0 \end{cases} \Rightarrow \begin{cases} aT^3 - \frac{3aT^3}{2} = \pi \\ b = -\frac{3aT}{2} \end{cases} \Rightarrow \begin{cases} -\frac{1}{2} a T^3 = \pi \Rightarrow a = -\frac{1}{T^3} 2\pi \\ b = \frac{6\pi}{2T^2} \end{cases}$$

$$\theta_d = -\frac{2\pi}{T^3} t^3 + \frac{6\pi}{2T^2} t^2 \quad \dot{\theta}_d = -\frac{6\pi}{T^3} t^2 + \frac{6\pi}{T^2} t \quad \ddot{\theta}_d = -\frac{12\pi}{T^3} t + \frac{6\pi}{T^2} = \frac{6\pi T - 12\pi t}{T^3}$$

$$u(t) = (I_c + m d^2) \left(-\frac{12\pi}{T^3} t + \frac{6\pi}{T^2} \right) + m g_0 \sin \left(-\frac{2\pi}{T^3} t^3 + \frac{6\pi}{2T^2} t^2 \right)$$

$$t = \frac{T}{2} \Rightarrow u\left(\frac{T}{2}\right) = m g_0 \leq u_{max} \text{ (sure)}$$

$$t = 0 \Rightarrow u(0) = (I_c + m d^2) \frac{6\pi}{T^2} \leq u_{max} \Rightarrow T^* = \sqrt{\frac{(I_c + m d^2) 6\pi}{u_{max}}}$$

Assume $T=1$ $(I_c + m d^2) 6\pi \geq u_{max}$ let κ such that:

$$(I_c + m d^2) \frac{6\pi}{\kappa^2} = u_{max} \Rightarrow \kappa = \sqrt{\frac{(I_c + m d^2) 6\pi}{u_{max}}}$$

Exercise 3

Figure 3 shows a simplified one-dimensional model of two robots permanently interacting in a compliant mode at the level of their end effectors¹. Compliance at the contact is modeled by a spring with stiffness $K > 0$. The two robots have equivalent masses m_1 and m_2 and are subject to control forces F_1 and F_2 . Their positions are given by q_1 and q_2 , with the zero reference for both variables corresponding to when the spring has no deformation (as shown in the figure).

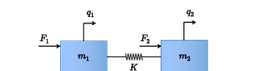


Figure 3: Two masses interacting through a spring.

Define the two control laws

$$F_1 = K_{P1} (q_{1d} - q_1) - K_{D1} \dot{q}_1, \quad F_2 = K_{P2} (q_{2d} - q_2) - K_{D2} \dot{q}_2, \quad (1)$$

with all gains strictly positive, and where the target positions q_{1d} and q_{2d} for the two masses are generic but different (i.e., $q_{1d} \neq q_{2d}$). These control laws have a decentralized structure, since they both use feedback information only local to the controlled mass, i.e., F_i is function only of $(q_i, \dot{q}_i)$, for $i = 1, 2$.

- Find the unique equilibrium state $(q, \dot{q}) = (\bar{q}, 0)$ for the closed-loop system under the control (1).
- Prove the global asymptotic stability of this equilibrium state by a Lyapunov/LaSalle argument.
- Is the equilibrium configuration $\bar{q}$ such that $\bar{q}_1 = q_{1d}$ and $\bar{q}_2 = q_{2d}$? If not, how would you modify the controllers (1), possibly keeping the decentralized structure, for the same previous target positions so that $q = q_d$ becomes the unique asymptotically stable equilibrium configuration?

i denote $m = m_1 = m_2$

the system is
$$\begin{cases} m \ddot{q}_1 + K(q_1 - q_2) = F_1 \\ m \ddot{q}_2 - K(q_1 - q_2) = F_2 \end{cases}$$
 closing the loop.

$$\begin{cases} m \ddot{q}_1 + K(q_1 - q_2) = K_{P1}(q_{1d} - q_1) - K_{D1} \dot{q}_1 \\ m \ddot{q}_2 - K(q_1 - q_2) = K_{P2}(q_{2d} - q_2) - K_{D2} \dot{q}_2 \end{cases} \Rightarrow \text{in the equilibrium i set } \dot{q}_1 = \dot{q}_2 = \ddot{q}_1 = \ddot{q}_2 = 0$$

$$\begin{cases} K(q_1 - q_2) = K_{P1}(q_{1d} - q_1) \\ K(q_2 - q_1) = K_{P2}(q_{2d} - q_2) \end{cases} \Rightarrow \begin{cases} (K + K_{P1})q_1 - Kq_2 = K_{P1}q_{1d} \\ (K + K_{P2})q_2 - Kq_1 = K_{P2}q_{2d} \end{cases} \Rightarrow \begin{cases} q_1 = (K + K_{P1})^{-1} (Kq_2 + K_{P1}q_{1d}) \\ q_2 = (K + K_{P2})^{-1} (Kq_1 + K_{P2}q_{2d}) \end{cases} \Rightarrow$$

$$q_2 = (K + K_{P2})^{-1} (K (K + K_{P1})^{-1} (K q_2 + K_{P1} q_{1d}) + K_{P2} q_{2d}) =$$

$$\frac{1}{K+K_{p2}} \left(\frac{K^2}{K+K_{p1}} q_2 + \frac{KK_{p1}}{K+K_{p1}} q_{1d} + K_{p2} q_{2d} \right) \Rightarrow$$

$$\left(1 - \frac{K^2}{K^2 + KK_{p1} + KK_{p2} + K_{p1}K_{p2}}\right) q_2 = \frac{1}{K + K_{p2}} \left(\frac{KK_{p1}}{K + K_{p1}} q_{1d} + K_{p2} q_{2d} \right) \Rightarrow$$

$$q_2 = \left(1 - \frac{K^2}{K^2 + KK_{P1} + KK_{P2} + K_{P1}K_{P2}} \right)^{-1} \frac{1}{K + K_{P2}} \left(\frac{KK_{P1}}{K + K_{P1}} q_{1d} + K_{P2} q_{2d} \right) = \frac{KK_{P1} q_{1d} + K_{P2} q_{2d} (K + K_{P1})}{K(K_{P1} + K_{P2}) + K_{P1}K_{P2}} = \bar{q}_2$$

$$\Rightarrow q_1 = \frac{K K_{p2} q_{10} + K_{p1} q_{1d} (K + K_{p2})}{K(K_{p1} + K_{p2}) + K_{p1} K_{p2}} = \bar{q}_1$$

This doesn't bring to the desired position, adding an integral term with suitable gain would stabilize the system at the desired configuration.

$$\begin{cases} m \ddot{q}_1 + K(q_1 - q_2) = K_{P1}(q_{1d} - q_1) - K_{D1} \dot{q}_1 \\ m \ddot{q}_2 - K(q_1 - q_2) = K_{P2}(q_{2d} - q_2) - K_{D2} \dot{q}_2 \end{cases} \quad \rightarrow \quad M \ddot{q} = \alpha - \hat{K} q - K_D \dot{q}$$

$$\Rightarrow \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix} \ddot{q} + \overbrace{\begin{pmatrix} K+K_{p1} & -K \\ -K & K+K_{p2} \end{pmatrix}}^{\tilde{K}} q + \overbrace{\begin{pmatrix} K_{D1} & 0 \\ 0 & K_{D2} \end{pmatrix}}^{K_D} \dot{q} - \overbrace{\begin{pmatrix} K_{p1} q_{1d} \\ K_{p2} q_{2d} \end{pmatrix}}^{u} = 0$$

We choose the following Lyapunov candidate: $e = \bar{q} - q$

$$V = \frac{1}{2} \dot{q}^T M \dot{q} + \frac{1}{2} e^T \hat{K} e \quad \text{is 0 only in } q = \bar{q}$$

$$\dot{V} = \dot{q}^T M \ddot{q} - e^T \hat{K} \dot{q} = \dot{q}^T (M \ddot{q} - \hat{K} e) = \dot{q}^T (\alpha - \hat{K} q - \kappa_0 \dot{q} - \hat{K} e) =$$

$$-\dot{\mathbf{q}}^T K_D \dot{\mathbf{q}} + \dot{\mathbf{q}}^T (\alpha - \hat{\mathbf{K}}(\mathbf{q} + \mathbf{e})) = -\dot{\mathbf{q}}^T K_D \dot{\mathbf{q}} + \underbrace{\dot{\mathbf{q}}^T (\alpha - \hat{\mathbf{K}} \bar{\mathbf{q}})}_0 = -\dot{\mathbf{q}}^T K_D \dot{\mathbf{q}} \leq 0$$

now let's check $\dot{V} = 0 \Rightarrow \dot{q} = 0$ $M\ddot{q} + \hat{K}q = \alpha \Rightarrow \ddot{q} = \hat{M}^{-1}(\alpha - \hat{K}q) = 0$

$$\Leftrightarrow q = \bar{q} \Rightarrow \bar{q} \text{ is GAS.} \quad \blacksquare$$